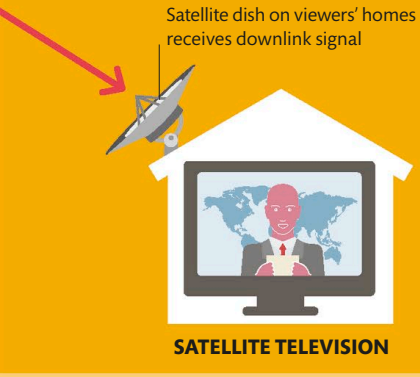
Analogue versus digital

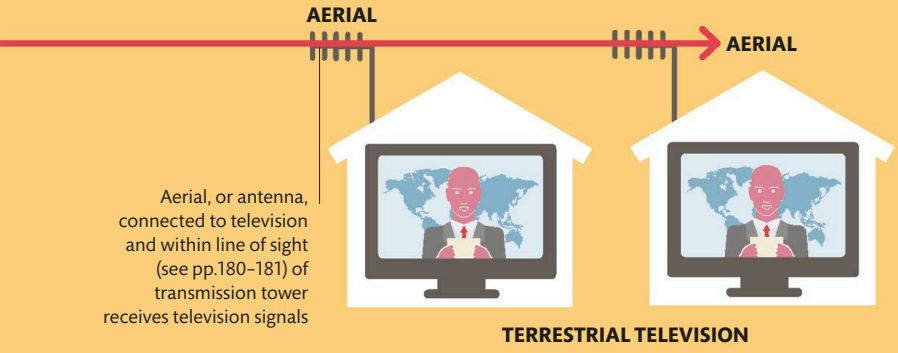
Broadcasters are in the process of switching entirely from analogue to digital television, which converts data into binary code before it is reassembled back to its original form. Digital television allows for improved image quality, more efficient use of the radio spectrum, and therefore a greater choice of channels than analogue television.

Analogue signal	Digital signal
Analogue signals vary continuously in frequency, amplitude, or both	Digital signals represent a series of pulses comprising just two states: on (1) or off (0)
Video quality is degraded during copying	Video quality does not change in copying
Uncompressed video wastes bandwidth	Compression allows more channels
Aspect ratio (screen width: height) is 4:3	More cinematic aspect ratio of 16:9
Much redundant information is transmitted	Only useful information is transmitted
Interference or "noise" is seen by viewer	Interference is suppressed



SATELLITE TELEVISION

WHEN THE SUN LINES UP BEHIND A SATELLITE, ITS MICROWAVE RAYS CAN DROWN OUT THE SIGNAL, CAUSING AN OUTAGE



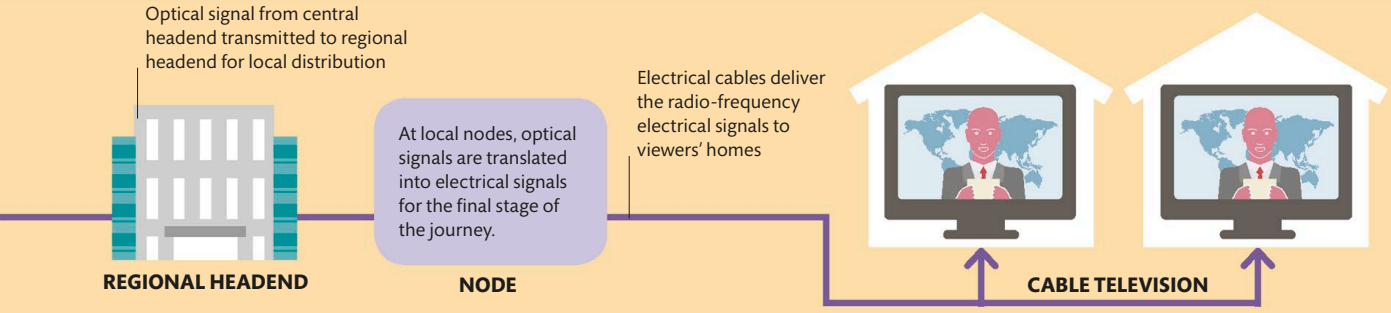
TERRESTRIAL TELEVISION

RECORDED TELEVISION

Videocassette recorders, which became popular in the 1980s, allowed viewers to record television programs on reels of magnetic tape to be played back later. Video is now almost always stored digitally. Today, much television programming is available on demand after, during, or without scheduled broadcast, meaning that viewers can stream programs online at their convenience.



ON-DEMAND SMART TV BOX



REGIONAL HEADEND

NODE

CABLE TELEVISION